3 . Stochastic gradient descent (SGD) with momentum

```
import tensorflow as tf
import matplotlib.pyplot as plt
import time

from tensorflow.keras import Sequentia
from tensorflow.keras.layers import F
from tensorflow.keras.datasets import

(x_train, y_train), (x_test, y_test) =

x_train = x_train.astype("float32") /
x_test = x_test.astype("float32") / 2

x_train = x_train[:20000]
y_train = y_train[:20000]

y_train = tf.keras.utils.to_categoric
y_test = tf.keras.utils.to_categorica

def create_model(optimizer):
    model = Sequential([
        Flatten(input_shape=(32, 32,
        Dense(256, activation="relu"),
        Dense(128, activation="relu"),
        Dense(10, activation="softmax
    ])
```

```
    model.compile(
        optimizer=optimizer,
        loss="categorical_crossentropy
        metrics=["accuracy"]
    )

    return model

sgd = tf.keras.optimizers.SGD(
    learning_rate=0.01,
    momentum=0.9
)

adam = tf.keras.optimizers.Adam(
    learning_rate=0.001
)

model_sgd = create_model(sgd)
model_adam = create_model(adam)

start = time.time()

history_sgd = model_sgd.fit(
    x_train,
    y_train,
    epochs=15,
    batch_size=128,
    validation_split=0.2,
    verbose=1
)
```

```
sgd_time = time.time() - start

start = time.time()

history_adam = model_adam.fit(
    x_train,
    y_train,
    epochs=15,
    batch_size=128,
    validation_split=0.2,
    verbose=1
)

adam_time = time.time() - start

sgd_loss, sgd_accuracy = model_sgd.eva
adam_loss, adam_accuracy = model_adam

print("SGD with Momentum")
print("Test Accuracy:", sgd_accuracy *
print("Test Loss:", sgd_loss)
print("Training Time:", sgd_time)

print("\nAdam Optimizer")
print("Test Accuracy:", adam_accuracy
print("Test Loss:", adam_loss)
print("Training Time:", adam_time)

plt.figure(figsize=(10, 5))
plt.plot(history_sgd.history["loss"],
plt.plot(history_adam.history["loss"]
plt.xlabel("Epoch")
```

```
plt.ylabel("Loss")
plt.title("Convergence Comparison - Lo
plt.legend()
plt.show()

plt.figure(figsize=(10, 5))
plt.plot(history_sgd.history["accuracy
plt.plot(history_adam.history["accurac
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.title("Convergence Comparison - Ac
plt.legend()
plt.show()

plt.figure(figsize=(10, 5))
plt.plot(history_sgd.history["val_loss
plt.plot(history_adam.history["val_los
plt.xlabel("Epoch")
plt.ylabel("Validation Loss")
plt.title("Validation Loss Comparison'
plt.legend()
plt.show()

plt.figure(figsize=(10, 5))
plt.plot(history_sgd.history["val_accu
plt.plot(history_adam.history["val_acc
plt.xlabel("Epoch")
plt.ylabel("Validation Accuracy")
plt.title("Validation Accuracy Compari
plt.legend()
plt.show()
```

```
Downloading data from https://www.cs.to
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━
/usr/local/lib/python3.13/dist-packages
  super().__init__(**kwargs)
Epoch 1/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 3s 16ms/st
Epoch 2/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/st
Epoch 3/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 3s 21ms/st
Epoch 4/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/st
Epoch 5/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 1s 11ms/st
Epoch 6/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 3s 13ms/st
Epoch 7/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/st
Epoch 8/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 3s 12ms/st
Epoch 9/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 15ms/st
Epoch 10/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 1s 11ms/st
Epoch 11/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 3s 12ms/st
Epoch 12/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 11ms/st
Epoch 13/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 1s 11ms/st
Epoch 14/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 3s 15ms/st
Epoch 15/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 1s 11ms/st
Epoch 1/15
```

```
125/125 ━━━━━━━━━━━━━━━━━━━━ 3s 15ms/st
Epoch 2/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/st
Epoch 3/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 3s 13ms/st
Epoch 4/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 18ms/st
Epoch 5/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 14ms/st
Epoch 6/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/st
Epoch 7/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/st
Epoch 8/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/st
Epoch 9/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 14ms/st
Epoch 10/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 17ms/st
Epoch 11/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/st
Epoch 12/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/st
Epoch 13/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/st
Epoch 14/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/st
Epoch 15/15
125/125 ━━━━━━━━━━━━━━━━━━━━ 2s 13ms/st
SGD with Momentum
Test Accuracy: 46.32000029087067
Test Loss: 1.529700279235839
Training Time: 32.339115619659424

Adam Optimizer
Test Accuracy: 44.89000141620636
```